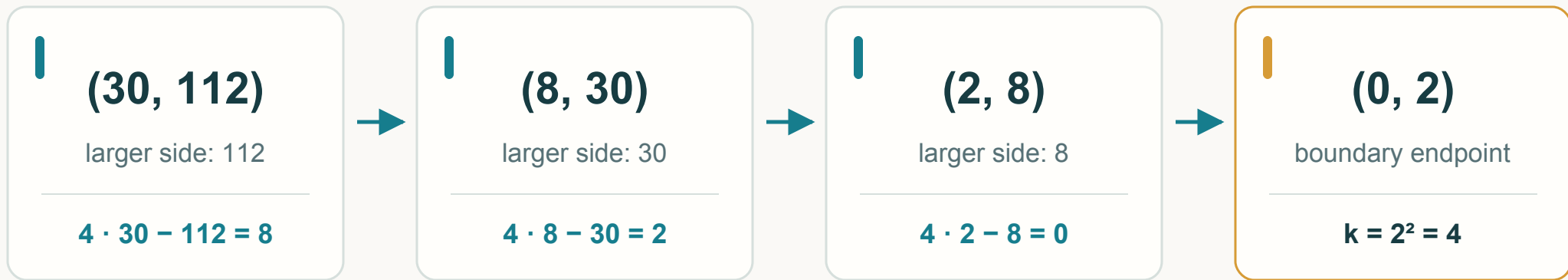


The same move repeats — until one side is zero

A complete descent for $k = 4$. This is a symbolic chain, not a length-scale diagram.



$$112 > 30 > 8 > 2$$

A strictly decreasing sequence of nonnegative integer sides cannot continue forever.